

A Slight (but extensive) Synopsis of Grade 9 Mathematics

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– *PART I.* –

*with questions + answer key at the end of the document.

INTRODUCTION TO QUADRATIC EQUATIONS

Mathematics is doable if you know mathematics.

Made with love and L^AT_EX.

Coded in Neovim with:

- VimTeX plugin with LaTeXmk Compiler and MikTeX for general L^AT_EX package installation and a local compiler.
- NvChad for theming based on IDE themes.
- Keyboard-Based PDF Reader **zathura**.
- *On Fedora 40 w/ KDE Plasma, by the way ;)*

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0.1 The Premise

In terms of learning, mathematics is a tough subject to ideally comprehend in terms on how readable or consistent the understanding is for the learner. It is normal for the regular learner to be overwhelmed or divided on certain definitions, or other uncertainties that the order of learning can not fully comprehend. However, this has become a problem for the task to aim a higher learning score on a text is regularly obliged to be necessary. It is come to a hard truth that a stereotype for mathematics is "the hard subject that is split between likeability and deduction." In terms of readability, the learner might know basic mathematics as the stepping-stone, but can not simply comprehend the ideal situations of using algebra in a useful sense. For example, if you were to learn the graph of a quadratic, would it needful to say it would be seen in a life where you (example only) work in a job that offers work within spreadsheets? Rather being, the readability is hard to comprehend as algebra contains letters, a complex web of manipulations, and dreadful formulas that is needed to fundamentally acquire the skill necessary for bigger scale syntax. The consistent understanding is harder to conquer, especially if the learner has troubling issues with learning mathematics.

However, my goal is to help (at the minimal scale if certain criteria are not met) aid the learner to consistently understand what most things are, and their certain manipulations. It is hard to understand mathematics if you are first learning it, and I do relate quite more than you originally think. If you still do not quite understand the concept, example, or real world application, then an aiding synopsis or summary can be useful and needless to say remarkably elegant and most likely impactful on understanding a whole. I plan to aid the reader, and hopefully fulfill the full understanding of an abstract picture with at least, some elegance sparkling. Of course, the whole picture might not be seen or made, and sometimes, even if the picture is fully made, no disclosure of unreadability, it might not be a detrimental picture to your life. But if you want a review, or a whole summary, an aid, or a synoptic lesson, then I hope to provide that. That picture is important, regardless of importance.

0.2 The Motivation

As a student, I formally regret not helping more clearly, more oftenly, and more actively. I want to make more clearer instructions, or more clearer definitions. I feel like some definitions are created dull, and some of mine are dull but with some sort of simplification of that dullness. I want to improve on that. I want to be more active and more oftenly show help to other people. It is very hard to explain, but if explaining the matter in any way is needful then I should show aid to those that really need the insight. Even though some other people may think I have done enough, I want to do more as a fulfillment. Ideally, my regrets are most often pardoned, but I appreciate my regrets as they can tell my weaknesses.

My motive to create this aiding factors in directly to my extensive motive of helping people. I am seen as someone who can easily understand the concept as my mathematics is highly regarded to be more skilled in my class. However, I do not wish to be egotistical of this title. I want to help others in a way that makes my soul glow in a dim abyss. My motive is directed to this, and I love doing it a lot. I do find it hard to explain, but help is help and explanations can be further extended eitherway. It is incredibly delighting me to show what I can explain to people as I want them to understand but also aim to a goal that they can set for them to easily see it as a puzzle piece with 16 pieces altogether. For that, this is why I wanted to make this.

0.3 "Motivational" Prologue

I understand completely that people fear mathematics. They think that some answers are weird, or that they can not derive to an answer that is reasonable to think. It is abominable for things like this to be prevented. They are mistakes. They are things that people have to experience in their daily lives. But mathematics comes from a large table of deduction and reasoning. If you make a mistake, do not brush it off as a form of 'lack of common sense' or 'something thereof I have forgotten' or anything that makes any mistake feel trivial and unconvoluted. However, it is most known that this is a bad habit. Make sense of that mistake. Of course, the occasional mishap of a leap of judgement is bound to happen. But be clear, it is not a reasonable excuse to blame it on a lack of thereof. It is more sensical to be seeing these mistakes as overestimation or underestimation. They may be too confident, or too scared to act on a proper point in thinking of an ideal next step. For that, it is normal. I know a lot of moments where a certain mishap occurs and it also tickles my knowledge. Take note that: If a mistake happens, it is normal. However, not every mistake is trivial.

Mistakes can happen. I've had a fair share, and it is quite embarrassing. However, this is not a sign that you are hopeless. Common mistakes are among everyone, rare mistakes are for an overlap of judgement or just something you never get, and perfect scores are just a way that you have avoided committing mistakes as a standpoint of a measurement for how undisputed your skill can be. It is understandable that you may feel that way, but others also know what it feels. Mistakes can happen to anyone, no matter the severity. My own self is filled with mistakes that I can either brush off and learn from it, or a piece of fact that I should adapt. It is normal and trivial for someone to experience mistakes. Perfection is never possible! You can have thousands of mistakes among an algebraic question, or even one on a pretty convoluted proof of a fundamental theorem. It is sensical to make one. If you think you wouldn't, it would be more likely that one can occur. Mathematics is like any ordinary task in your life (seen from a standpoint, let's say.) If you are bound to make a mistake, it will of course happen, but learn from it. Learn why you always see this mistake in your exams. Learn why you identified the letter wrong. Learn why you did not read instructions. They happen! Of course, repeating them is not healthy. All you need to do is worry about if you are going to make a mistake (but not with over-pessimism.) Think about it, and

carefully take your time but make it quick. Do not consider yourself inferior for a simple mistake you made, it is just people who are superior just learnt from their mistakes. You won't get it on your first try, but that's the reason why it is a simple "try." Try again, and see if you will get it. If not, repeat the step. But if it doesn't even after all, then consider it a footnote for you to think later.

NOTES OF APPRECIATION:

*My classmates,
And my mathematics teacher*

Chapter 1

Prerequisites

The topics covered in Grade 9 Mathematics is extensive on knowledge on how to solve equations, the least common denominator, and a lot more. This chapter aims to provide at least the most essential prerequisite topics that is known to come in the following chapters.

We will assume the learner knows basic algebra, namely:

- Arithmetic With Integers
- Arithmetic With Fractions
- Difference Between Expressions and Equations
- Variables, Coefficients, and Constants
- Basics of Equality
- Basic Algebraic Properties Among Arithmetic Operations
 1. Associative Property
 2. Closure Property
 3. Commutative Property
 4. Distributive Property
 5. Identity Property
 6. Inverse Property
 7. Zero Property
- Exponents and Radicals
- Familiarity with Like and Unlike Terms
- Cancelling and Simplifying Terms
- Product/Quotient Rule for Exponents (Radicals is not necessary.)
- Operations with Like Terms (Basic Monomials)

If the learner is not familiar with all of the above, you may take your moment and review or read more about what they are. The prerequisite topics that I will try to discuss in a manner that is understandable are the most barebone-essential parts of this topic. It may be optional to review more, but is not recommended as other topics might interfere with the original goal. These prerequisites will also apply to Parts II-IV, as well with the prerequisites with Parts II-IV included. We shall begin.

1.1 Polynomials

A *polynomial* is an algebraic equation that has constants, coefficients, and variables, but does not involve division of a variable, variables with negative exponents, variables with exponents with fractions, or any variable under a radical. It should also not be infinite. We consider expressions that follow the criteria polynomials, but the following are polynomials as equations.

Example 1.1.1 (Examples of Polynomials).

$$x^2 - 100x + 19 = 30$$

$$x + 2 = 1$$

$$x^5 - 10x^2 + 1 = 39$$

$$x - 20 = -14x$$

$$-1 = x^3 + 41$$

$$x^2 = 0$$

$$x + 52 = 13x^4 + 2x^2$$

$$12 = 12$$

All of these polynomials are in single variable. We will consider our cases of polynomials to be single variable because double or triple variables involves the idea of systems of equations, which is not within our range.

1.1.1 Degree of Polynomials

The *degree* of a polynomial is the highest possible exponent that exists in a polynomial. The degree of any equation is the highest possible exponent, but we are of course talking about polynomials.

Example 1.1.2 (Degrees of Sample Polynomials). The following.

- 1) $x^3 + 15x - 100$ has the degree of 3. The highest power is x^3 .
- 2) $x + 11$ has the degree of 1. The highest power is x^1 which is x itself.
- 3) 3 has the degree of 0. This is a special case, as all coefficients are zero, or there is a hidden $3(x^0)$ wherein $x^0 = 1$.

1.1.2 Kinds of Polynomials

Polynomials are characterized to be in many forms. This topic is quite short, so I will try to keep it brief:

1. A *monomial* is a polynomial that has ONLY one term. $5x$, x^2 , or 12 are monomials.
2. A *binomial* is a polynomial that has ONLY two terms. $x^2 + 5$, $x^3 - 6$, or $x - 7$ are binomials.
3. A *trinomial* is a polynomial that has ONLY three terms. $x^4 + 4x^2 + 15$, $x^5 + x + 29$, or $x^2 + x + 2$ are trinomials.

Another kind of polynomials are based off their degree.

1. A *linear polynomial* is a polynomial that has the degree of one.

2. A *quadratic polynomial* is a polynomial that has the degree of two.
3. A *cubic polynomial* is a polynomial that has the degree of three.
4. A *quartic polynomial* is a polynomial that has the degree of four.
 - A quartic polynomial that has even exponents like $2x^4 + 12x^2 - 15$ is called a *biquadratic polynomial*.
5. A *quintic polynomial* is a polynomial that has the degree of five.

There are more kinds of polynomials that are based off a degree. However, we are mostly more focused on degree two and four kinds of polynomials later on in the future chapter nextward. We are *only* considering single-variable polynomials. Those with double variables are for future topics.

1.2 Operations on Polynomials

Operations on polynomials are needful in the topics foreshadowing. To give a gist, the following can be shown:

1.2.1 Adding/Subtracting Polynomials

Adding polynomials is like adding regular like terms. However, we must also perform the distributive property if we are subtracting polynomials. Some examples we can show:

Example 1.2.1 (Addition I.). Solve $(3x^2 + 15x - 2) + (x^2 - 11)$.

$$\begin{aligned}
 (3x^2 + 15x - 2) + (x^2 - 11) &= 3x^2 + 15x - 2 + x^2 - 11 \\
 (3x^2 + x^2) + (-2 - 11) + 15x &= 4x^2 - 13 + 15x \\
 &= \boxed{4x^2 + 15x - 13}
 \end{aligned}$$

Example 1.2.2 (Addition II.). Solve $(x + 15) + (14x - 22)$.

$$\begin{aligned}
 (x + 15) + (-14x - 22) &= x + 15 - 14x - 22 \\
 (x - 14x) + (15 - 22) &= -13x - 7 \\
 -(-13x - 7) &= \boxed{13x + 7}
 \end{aligned}$$

The reason why we multiply it with a negative is because it is unconventional to have a negative term with the highest power in the polynomial, if you are sorting by descending order (the most popular kind.)

Example 1.2.3 (Subtraction I.). Solve $(14x^2 + x - 20) - (-2x^2 + 5x - 9)$.

$$\begin{aligned}
 (14x^2 + x - 20) - (-2x^2 + 5x - 9) &= 14x^2 + x - 20 + (-1)(-2x^2 + 5x - 9) \\
 14x^2 + x - 20 + 2x^2 - 5x + 9 &= (14x^2 + 2x^2) + (x - 5x) + (-20 - 9) \\
 &= \boxed{16x^2 - 4x - 29}
 \end{aligned}$$

Example 1.2.4 (Subtraction II.). Solve $(x^2 - 2x + 6) - (5x^2 + 6x)$.

$$\begin{aligned}
 (x^2 - 2x + 6) - (5x^2 + 6x) &= x^2 - 2x + 6 + (-1)(5x^2 + 6x) \\
 x^2 - 2x + 6 - 5x^2 - 6x &= (x^2 - 5x^2) + (-2x - 6x) + 6 \\
 -4x^2 - 8x + 6 &= -(-4x^2 - 8x + 6) \\
 &= \boxed{4x^2 + 8x - 6}
 \end{aligned}$$

1.2.2 Multiplying/Dividing Polynomials

The learner was assumed to have known the distributive property. This applies to all polynomials when performing multiplication. However, multiplying two binomials together can be simplified by using *FOIL*. It is a methodical approach to multiplying two binomials and a more simplified version of the distributive property for the product of two binomials.

Definition 1.2.1 (The FOIL Method). If $(a + b)(c + d)$ exists, the methodical approach of FOIL stands for - **F**irst, **I**nnner, **O**uter, **L**ast. In the product:

$$\begin{aligned} F &= ac \\ O &= ad \\ I &= bc \\ L &= bd \\ &= ac + ad + bc + bd \\ &= (a + b)(c + d) \end{aligned}$$

This approach is only for multiplying two binomials together. Other methods require the distributive property.

Example 1.2.5 (Multiplication I.). Solve $(2x + 1)(x^2 - 5)$. Using FOIL:

$$\begin{aligned} F &= (2x)(x^2) = 2x^3 \\ O &= (2x)(-5) = -10x \\ I &= (1)(x^2) = x^2 \\ L &= (1)(-5) = -5 \end{aligned}$$

Add them together:

$$2x^3 + (-10x) + x^2 + (-5) = \boxed{2x^3 + x^2 - 10x - 5}$$

With the Distributive property:

$$\begin{aligned} (2x + 1)(x^2 - 5) &= 2x(x^2 - 5) + 1(x^2 - 5) \\ 2x(x^2) - 2x(5) + x^2 - 5 &= 2x^3 + x^2 - 10x - 5 \end{aligned}$$

Example 1.2.6 (Multiplication II.). Solve $(x - 5)(x^2 + 2x - 7)$. Since this is not a product of two binomials, we shall use the distributive property.

$$\begin{aligned} (x - 5)(x^2 + 2x - 7) &= x(x^2 + 2x - 7) + (-5)(x^2 + 2x - 7) \\ x^3 + 2x^2 - 7x - 5x^2 - 10x - 14 &= x^3 + (2x^2 - 5x^2) + (-7x - 10x) - 14 \\ &= \boxed{x^3 - 3x^2 - 17x - 14} \end{aligned}$$

To divide polynomials, you need to do the following steps. It is a tough process that took me a year to finally understand:

Definition 1.2.2 (How to Divide Polynomials). To divide two polynomials in LONG division:

1. Divide the first term of the polynomial you want to divide to the other polynomial's first term.

We continue the process like normal. By the way, the macro I am using to compute the long division has been modified such that the long division is proper with missing terms. Just omit the final row, as $0x + 0$ is just 0. We just repeat the same steps. However, we just note that the zeroes are simply there to show how the process is carried. Again, the macro does not omit the products of zero (in the solution) and is a little messy. However, if we just acknowledge we can omit the zeroes, we can still get a defined result of $x^2 + 5$.

However, we know for a fact that the idea of a *remainder* exist in long division. So, if there exists a remainder, then the quotient is represented as "POLYNOMIAL + (REMAINDER)/(DIVISOR)".

Example 1.2.9 (Division III.). Solve $\frac{2x^2 + 5x - 3}{x - 2}$.

$$\begin{array}{r} 2x + 9 \\ x - 2 \overline{) 2x^2 + 5x - 3} \\ \underline{- 2x^2 + 4x} \\ 9x - 3 \\ \underline{- 9x + 18} \\ 15 \end{array}$$

$$= 2x + 9 + \frac{15}{x-2}$$

We will not accept simplifying the entire fraction to be negative if either numerator or denominator (assuming both are not the same sign) is negative, as personally, it just confuses me to explain it.

1.3 Factoring Polynomials

Factoring is the idea of simplifying a polynomial as a product (if possible.) For example:

Example 1.3.1.

$$4x + 12 = 4(x + 3)$$

$$x^2 - y^2 = (x + y)(x - y)$$

$$a^3 + 8 = (a + 2)(a^2 + 2a + 4)$$

The kinds of factoring include:

1. Factoring by GCF (Common Monomial Factoring)
2. Difference of Two Squares
3. Perfect-Square Trinomial
4. Sum/Difference of Two Cubes

Really, the sum/difference of two cubes is not really related to degree-two polynomials since they do not appear (mostly) in the proceeding. However, it is still worthy to mention as a refresher.

1.3.1 Factoring by GCF

The *Greatest Common Factor* is the greatest factor that exists between two terms. For example, the GCF of 100 and 75 is 25. Reasoning is, we can divide 100 and 75 by 25, as well as 25 is the biggest number that can divide both 100 and 75.

The GCF of similar variables with varying exponents is always the smallest exponent. For example, the GCF of x^2 , x^7 , and x^4 is x^2 because it is the greatest exponent that can divide all three. The GCF of $15x^3$, $3x$, and $9x^4$ is $3x$ because $3x$ divides to all three, and 3 is the GCF of all the coefficients and x^2 is the GCF of all the variables in terms of x .

This information is useful because the idea of *Factoring by GCF* or by book-terms, *common monomial factoring* is taking the GCF of all the terms in a polynomial. If all of them have a common factor greater than 1, we divide all the terms by the GCF and make the GCF and the divided polynomial a product. I know it sounds wordy, but the following may help:

Example 1.3.2. Factor $4x^2 - 120$.

The GCF of all of them is 4, of course. We know that x only is on 4, hence the GCF of x is 1. We check the coefficients, and therefore we know that 4 and 120 can divide 4 easily.

$$\begin{aligned} 4x^2 - 120 &= \frac{4x^2}{4} - \frac{120}{4} \\ x^2 - 30 &= 4(x^2 - 30) \end{aligned}$$

We need to show how it becomes the polynomial beforehand. So we always put the GCF and the factored form of the polynomial as a product. As we know, the GCF is 4, and our factored form is $x^2 - 30$ if we divide it by our GCF. Put them both as a product. We can always check this by using the distributive property:

$$\begin{aligned} 4(x^2 - 30) &= 4(x^2) - 4(30) \\ &= 4x^2 - 120 \end{aligned}$$

Example 1.3.3. Factor $8x^3 - 16x^2 + 12x$.

$$\begin{aligned} 8x^3 - 16x^2 + 12x &= \frac{8x^3}{4x} - \frac{16x^2}{4x} + \frac{12x}{4x} \\ 5x^2 - 4x + 3 &= 4x(5x^2 - 4x + 3) \end{aligned}$$

We know that our GCF is $4x$ because $4x$ divides all the terms in the polynomial. Divide all the terms, and make the factored polynomial and the GCF into a product. We get our final answer.

TASK 1

Determine if $4x(5x^2 - 4x + 3) = 8x^3 - 16x^2 + 12x$.

NOTE: Importantly, if you see anything that looks like it can be factored, try to see if it is factorable. It is common notion to always seek patterns in polynomials especially when trying to cancel terms.

We now know the ideas for Factoring by GCF. Ideally, we are only going to focus on single-variable polynomials to not overwhelm the reader. You are, however, encouraged to do so if you have not taken 8th Grade Mathematics before.

1.3.2 The Difference of Two Squares

The *difference of two squares* is the following:

Definition 1.3.1 (Difference of Two Squares). If x and y are as $x^2 - y^2$, then $x^2 - y^2 = (x+y)(x-y)$.

If two numbers x and y are squares, then they can be factored accordingly. We must note, however, that, sometimes, either x or y (mostly y) are not perfect squares. So, we must always involve the square root.

Remark. $(\sqrt{a})^2 = a$. It always cancels the root. As well, the root of a is always the solution to a^2 , for if a is an integer.¹

We will provide three examples of perfect-squares and two examples of irrational factorizations (which generally are not acceptable but for our means it is for this subsection.)

Example 1.3.4. Factor $x^2 - 25$.

$$\begin{aligned} x^2 - 25 &= (x)^2 - (5)^2 \\ &= (x+5)(x-5) \end{aligned}$$

We must know that $5^2 = 25$ because $5 \cdot 5$. I don't really need to explain for x^2 .

Example 1.3.5. Factor $b^2 - 1$.

$$\begin{aligned} b^2 - 1 &= (b)^2 - (1)^2 \\ &= (b+1)(b-1) \end{aligned}$$

1^2 is trivial. I mean...

Example 1.3.6. Factor $\frac{4}{9}x^2 - 49$.

$$\begin{aligned} \frac{4}{9}x^2 - 49 &= \left(\frac{2}{3}x\right)^2 - (7)^2 \\ &= \left(\frac{2}{3}x + 7\right)\left(\frac{2}{3}x - 7\right) \end{aligned}$$

Example 1.3.7. Factor $p^2 - 7$.

We have come to realize that 7 is not a perfect square. From our Remark earlier, we know that $(\sqrt{7})^2 = 7$. So we must know that $\sqrt{7}$ is a solution for 7 represented as a perfect square. Therefore:

$$\begin{aligned} p^2 - 7 &= (p)^2 - (\sqrt{7})^2 \\ &= (p + \sqrt{7})(p - \sqrt{7}) \end{aligned}$$

Example 1.3.8. Factor $x^2 - 17$. Using our Remark:

$$\begin{aligned} x^2 - 17 &= (x)^2 - (\sqrt{17})^2 \\ &= (x + \sqrt{17})(x - \sqrt{17}) \end{aligned}$$

¹This will be explained more further when we go to quadratic equations. For now, acknowledge the fact that this is true.

More further about this is in the future topic.

TASK 2

Show that, from **Example 1.3.8**, $(x + \sqrt{17})(x - \sqrt{17}) = x^2 - 17$ by using FOIL.

HINT: Use the remark.

1.3.3 Perfect-Square Trinomial

Definition 1.3.2 (Perfect-Square Trinomial). If x and y are as $x^2 + 2xy + y^2$, then $x^2 + 2xy + y^2 = (x + y)^2$. Similarly, if x and y are as $x^2 - 2xy + y^2$, then $x^2 - 2xy + y^2 = (x - y)^2$.

If we just remember the pattern of $x^2 + 2xy + y^2$ or $x^2 - 2xy + y^2$ wherein x and y are two different numbers, and their factored forms, then factoring the pattern of such is useful. Needful to say that perfect-square trinomials are going to be found in solving quadratics, so be noted of such.

Example 1.3.9. Factor $x^2 + 4x + 4$.

$$\begin{aligned} x^2 + 4x + 4 &= (x)^2 + (2)(2)(x) + (2)^2 \\ &= (x + 2)^2 \end{aligned}$$

We can use FOIL to prove this statement:

$$\begin{aligned} (x + 2)^2 &= (x + 2)(x + 2) \\ F &= x \cdot x \\ O &= x \cdot 2 \\ I &= 2 \cdot x \\ L &= 2 \cdot 2 \end{aligned}$$

We get:

$$\begin{aligned} &= x^2 + 2x + 2x + 4 \\ &= x^2 + 4x + 4 \end{aligned}$$

Example 1.3.10. Factor $9x^2 - 24x + 16$

$$\begin{aligned} 9x^2 - 24x + 16 &= (3x)^2 - 2(3x)(4) + (4)^2 \\ &= (3x - 4)^2 \end{aligned}$$

Example 1.3.11. Factor $\frac{x^2}{4} - 3x + 9$.

$$\begin{aligned} \frac{x^2}{4} - 3x + 9 &= \frac{1}{4}x^2 - 3x + 9 \\ \left(\frac{1}{2}x\right)^2 - 2(3)\left(\frac{1}{2}x\right) + (3)^2 &= \left(\frac{1}{2}x - 3\right)^2 \\ &= \left(\frac{x}{2} - 3\right)^2 \end{aligned}$$

In **Example 1.3.11**, we note that multiplying an integer by a fraction is the same as that integer in the numerator of the fraction, being multiplied by the same denominator e.g. $\frac{6}{4} = \frac{2}{4} \cdot 2$, $\frac{4x}{8} = \frac{1}{8} \cdot 4x$ etc. This is explained in the next subsection.

1.3.4 The Sum & Difference of Two Cubes

Definition 1.3.3 (Sum of Two Cubes). If x and y are as $x^3 + y^3$, then $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$.

Definition 1.3.4 (Difference of Two Cubes). If x and y are as $x^3 - y^3$, then (next line)
 $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$.

Pretty straightforward. Like usual, identify a pattern if it is simplifiable by the sum and difference of two cubes.

Example 1.3.12. Factor $8x^3 - 125$.

$$\begin{aligned} 8x^3 - 125 &= (2x)^3 - (5)^3 \\ (2x - 3)((2x)^2 + (2x)(3) + (3)^2) &= (2x - 3)(4x^2 + 6x + 9) \end{aligned}$$

Example 1.3.13. Factor $27x^3 + 216$.

$$\begin{aligned} 27x^3 + 216 &= (3x)^3 + (6)^3 \\ (3x + 6)((3x)^2 - (3x)(6) + (6)^2) &= (3x + 6)(9x^2 - 18x + 36) \end{aligned}$$

1.3.5 Patterns in Factoring

Factoring a polynomial completely relies on decisive thinking on what pattern is familiar. For example, a polynomial can have the form that is not factorable with the difference of two squares, but in reality, we just need to perform Common Monomial Factoring first beforehand then factor by the difference of two squares.

Example 1.3.14. Factor $12x^3 - 48x$.

Start with CMF:

$$\begin{aligned} 12x^3 - 48 &= \frac{12x^3}{12x} - \frac{48x}{12x} \text{ for } 12x \text{ is their GCF} \\ x^2 - 4 &= 12x(x^2 - 4) \end{aligned}$$

Then factor the inside using the Difference of Two Squares:

$$\begin{aligned} 12x(x^2 - 4) &= 12x((x)^2 - (2)^2) \\ &= 12x((x + 2)(x - 2)) \end{aligned}$$

Example 1.3.15. Factor $5x^2 - 30x + 45$.

Start with CMF:

$$\begin{aligned} 5x^2 - 30x + 45 &= \frac{5x^2}{5} - \frac{30x}{5} + \frac{45}{5} \\ x^2 - 6x + 9 &= 5(x^2 - 6x + 9) \end{aligned}$$

Then factor the inside using the Perfect-Square Trinomial:

$$\begin{aligned} 5(x^2 - 6x + 9) &= 5((x)^2 - 2(3)(x) + (3)^2) \\ &= 5(x - 3)^2 \end{aligned}$$

Example 1.3.16. Factor $16x^6 + 250x^3$.

Start with CMF:

$$\begin{aligned} 16x^6 + 250x^3 &= \frac{16x^6}{2x^3} + \frac{250x^3}{2x^3} \\ 8x^3 + 125 &= 2x^3(8x^3 + 125) \end{aligned}$$

Then factor the inside using the Sum of Two Cubes:

$$\begin{aligned} 2x^3(8x^3 + 125) &= 2x^3((2x)^3 + (5)^3) \\ 2x^3((2x + 5)((2x)^2 - (2x)(5) + (5^2))) &= 2x^3((2x + 5)(4x^2 - 10x + 25)) \end{aligned}$$

1.3.6 Factoring Trinomials

Trinomials involve three terms. They can be factored, however, they involve special patterns unlike so. We shall start by the following kind of factoring:

Definition 1.3.5 (Factoring Trinomials I.). Let a polynomial $ax^2 + bx + c$ exist. If $a = 1$, then $x^2 + bx + c$ is the new form. Let $b = m + n$ and $c = mn$. Then $x^2 + (m + n)x + mn$ exists.

$$\begin{aligned} x^2 + (m + n)x + mn &= x^2 + mx + nx + mn \\ (x^2 + mx) + (nx + mn) &= x(x + m) + n(x + m) \\ &= (x + m)(x + n) \end{aligned}$$

Therefore $(x + m)(x + n)$ is the factor of $x^2 + bx + c$ considering $b = m + n$ and $c = mn$

I really went overboard. Okay:

SIMPLIFICATION: If two numbers m and n are two numbers that add up to b and multiply to get c in $x^2 + bx + c$, the factored form is $(x + m)(x + n)$.

Remark. Note that: Since m and n multiply to get c , m and n must be factors of c .

Let us try an example:

Example 1.3.17. Factor $x^2 + 5x + 6$.

Using our remark, we know that the factors of 6 are 1, 2, 3, 6. We can tell that $2 + 3 = 5$ and 2 and 3 are factors of 6. Therefore, we can tell that the factored form of this polynomial is $(x + 2)(x + 3)$. We can prove this by FOIL or the distributive property:

$$\begin{aligned} (x + 2)(x + 3) &= x(x + 3) + 2(x + 3) \\ x^2 + 3x + 2x + 6 &= x^2 + 5x + 6 \end{aligned}$$

We must note that: c can of course be in different signs. We can emerge a pattern:

Note. If $c > 0$, then either m and n are both positive or negative. If $c < 0$, then either m is negative and n is positive or vice versa (m and n have opposing signs.)

Remark. If $c > 0$, then either b is negative or positive BECAUSE either m and n is both positive or negative. If $c < 0$, then either b is negative or positive DEPENDENT on if the sum of m and n is negative or positive.

To get what I mean, let us try four examples:

Example 1.3.18. Factor $x^2 - x - 6$.

Since trivially $-1 < 0$ for b , we must know that m and n are opposing signs. For c , we can try $(-2)(3)$. However, $-2 + 3 = 1$, and b is not $+1$. However, $(-3)(2)$ is a viable solution as $-3 + 2 = -1$. Hence we know what m and n are. Our factored form therefore must be $(x - 3)(x + 2)$.

Example 1.3.19. Factor $x^2 - 8x + 15$.

Since trivially $15 > 0$, we must know that m and n are similar signs noting that $b = -8$. m and n must be both negative. Trying $(-5)(-3) = 15$, and $-5 + (-3) = -8$, we can now say that the factored form is $(x - 5)(x - 3)$.

So far straightforward. If you need a moment, you can reread.

Example 1.3.20. Factor $x^2 + 3x - 10$.

Since trivially $-10 < 0$, and $3 > 0$, we must know that m and n are opposing signs. We can try $(-5)(2)$ but it won't add up to $+3$. But $(-2)(5)$ does add up to $+3$. We can safely say that the factored form is $(x - 2)(x + 5)$.

Example 1.3.21. Factor $x^2 - 14x + 49$.

Since trivially $49 > 0$, and $-14 < 0$, we must know that m and n are similar signs. We can easily tell that $(-7)(-7)$ is a factor that adds to -14 and multiplies to 49 . We can say that our factored form is $(x - 7)(x - 7)$ OR $(x - 7)^2$.

As you can see in **Example 1.3.21**, our factored form results to a perfect-square trinomial. This is one of the methods to show how a perfect-square trinomial forms.

1.4 Rational Expressions

A *rational expression* is simply just a fraction, but normally is with spooky algebra stuff e.g:

Example 1.4.1.

$$\frac{\frac{12x^2}{5}}{\frac{15x+3}{2x} \cdot \frac{7}{11}}$$

Again, a rational expression is just a fancy way to say that it's a fraction. I must assume that you know how cancelling terms works. It is just cancelling like terms among the numerator and the denominator as something divided by itself is always 1. You can research more about it if you are unfamiliar with such.

1.4.1 Multiplying and Dividing Expressions

A review: If you know how to multiply fractions, they are just straightforward.

Definition 1.4.1 (Multiplying Fractions).

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

And dividing fractions is also straightforward but with extra steps.

Commonly coined as "Keep - Change - Flip," methodically, it is: *Keep* the first fraction like normal, *change* the division sign to the multiplication sign, and *flip* the other fraction switching the denominator with the numerator. "Flipping" a fraction is the process of *reciprocating*. A flipped fraction is a *reciprocal* of the original fraction. Formula-wise:

Definition 1.4.2 (Dividing Fractions).

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c} = \frac{ad}{bc}$$